

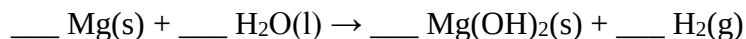
Team Number: _____

PROBLEM 1

10% of total

TOTAL	A	B	C	D	E	F
16	2	2	3	3	4	2

A strip of magnesium ribbon is dropped into hot water, and the following reaction takes place:



- Balance the reaction by filling in the coefficients, using lowest whole numbers.
- Why must the magnesium ribbon be dropped into hot water to induce a reaction?
- After a period of time, the reaction slows and the evolution of bubbles ceases. Explain these observations. Hint: Think about solubility.
- Assuming that the magnesium reacts completely with the water, calculate the mass of H_2 formed when 0.420g of magnesium ribbon is introduced.

Mass of H_2 =

- The hydrogen gas evolved is captured and combusted with 0.250g of oxygen. If 0.230g of water are produced, find the limiting reagent and calculate the % yield of water.

Limiting Reagent:

% Yield =

- When magnesium is reacted with hot steam, it reacts with it in a 1:1 ratio to form two products. Write the equation of the balanced reaction.

Balanced Equation:

Team Number: _____

PROBLEM 2

11% of total

TOTAL	A	B	C	D	E	F
18	5	3	2	2	4	2

An unknown metal X from the Group 1 elements (alkali metals) burns in a pure oxygen environment with an intense red flame to form a white compound.

a. Write the product of combustion (the metal oxide) that predominates for each the alkali metals, from lithium to cesium.

Li:

Rb:

Na:

Cs:

K:

b. Assuming that 0.200g of the unknown metal X forms 0.431g of the oxide, identify X.

Identity of X:

c. When X is burned in atmospheric air, the metal oxide is not the exclusive product. The metal also reacts with nitrogen to form a metal nitride product. Write down the balanced equation of this nitrogen reaction (use X in place of the metal if you could not determine its identity).

Balanced Equation:

d. The metal nitride reacts with water to form a gaseous product. Identify this gas.

Identity of Gas:

e. 0.300g of metal X are burned in atmospheric air to form 0.600g of a mixture of the metal oxide and nitride. Find the mass of each product formed.

Mass of Metal Oxide =

Mass of Metal Nitride =

f. Metal X is treated with hydrogen gas to form an ionic compound. Identify this ionic compound.

Identity of X:

Team Number: _____

PROBLEM 3

11% of total

TOTAL	A	B	C	D	E	F	G	H
18	2	2	2	2	2	4	2	2

Write down the net ionic equation for each question below (*does not need to be balanced*). Also classify the reactions as being combustion, acid-base, organic, decomposition, combination, precipitation, nuclear, oxidation-reduction, or complex-ion formation. Multiple reaction types may apply, but you need only write one.

- Solutions of sodium iodide and lead nitrate are mixed.
- Carbon dioxide is bubbled through a solution of calcium hydroxide.
- A solution of hydrofluoric acid is added to a suspension of aluminum hydroxide.
- Phosphorus is burned in excess oxygen.
- Uranium-238 undergoes alpha decay.
- $\text{YBa}_2\text{Cu}_3\text{O}_7$ is dissolved in aqueous HCl with excess KI in an oxygen-free atmosphere.
- Concentrated hydrochloric acid is added to a solution of cobalt(II) chloride and heated.
- Acetic acid is heated with ethanol in the presence of H^+

Team Number: _____

PROBLEM 4

7% of total

TOTAL	A	B	C	D	E
12	3	3	2	2	2

Breathalyzers make an estimate of blood alcohol by measuring the amount of alcohol in a person's breath. Old breathalyzer models functioned by having a person breath through a solution of potassium dichromate, which oxidizes the ethanol in the person's breath into acetic acid. The dichromate changes color in the process, which is measured by a detector.

a. Balance each of the equations below for the oxidation of ethanol and reduction of dichromate.



b. Combine the two equations to write a net ionic equation.

Equation:

The concentration of ethanol in the person's blood to ethanol in the person's breath is on average 2100. Thus, this is the standard most machines are calibrated to. The current legal limit on driving in the United States is 0.08g ethanol per 1000cm³ of blood.

c. If a person at the limit exhales 1L of breath, how many moles of ethanol are breathed out?

Moles of Ethanol =

d. If 14% of the dichromate molecules in the breathalyzer have reacted, how many molecules of dichromate does the breathalyzer in total contain?

Molecules of Dichromate =

e. A person's ratio of ethanol in blood to ethanol in breath is 2300. The BAC reported by the breathalyzer will be: *Circle the correct choice*

Higher than the real BAC

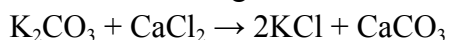
Lower than the real BAC

CHEMICAL REACTIONS & STOICHIOMETRY (30 points)

- Which of the following most accurately defines a “limiting reactant”?
 - the reactant that is present in excess and determines the amount of product formed
 - the reactant that is completely consumed in a chemical reaction
 - the reactant that remains unchanged after the reaction
 - the reactant that is used in the highest quantity during the reaction
- 1.0 M HCl is used to titrate a solution of KOH, AgNO₃, and BaBr₂. After titration is complete, a powder is observed in the reaction flask. What is the likely identity of this precipitate?
 - AgBa
 - HNO₃
 - AgCl
 - BaCl₂
- K₂S undergoes a gas evolution reaction: $\text{K}_2\text{S}(\text{aq}) + 2\text{HCl}(\text{aq}) \rightarrow \text{H}_2\text{S}(\text{g}) + 2\text{KCl}(\text{aq})$. After completing this reaction in a lab at STP, the gas pressure in the reaction flask increased by 1.00 atm. How many grams of K₂S must have reacted in order for this effect to be observed? (M = 110.262 g/mol K₂S)
 - 4.922 g K₂S
 - 55.131 g K₂S
 - 110.262 g K₂S
 - 220.524 g K₂S
- Silicon reacts with water to produce silica (SiO₂). If the rate of disappearance of water is 0.050 moles/minute, then what is the rate of production of silica?
 - 0.017 moles/minute
 - 0.025 moles/minute
 - 0.050 moles/minute
 - 0.100 moles/minute
- How many moles of Na₂SO₄ are present in 50.0 grams of Glauber salt (Na₂SO₄•10H₂O)? (M = 142.04 g/mol Na₂SO₄, 18.02 g/mol H₂O)
 - 0.1552 mol Na₂SO₄
 - 0.1968 mol Na₂SO₄
 - 0.3124 mol Na₂SO₄
 - 0.4408 mol Na₂SO₄
- Which of the following hydrocarbon fuels consumes the most moles of O₂ per mole of fuel combusted?
 - C₅H₁₂
 - C₆H₁₂
 - C₆H₁₄
 - C₇H₈

7. Consider the reaction: $\text{MgCl}_2(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{MgSO}_4(\text{aq}) + 2\text{HCl}(\text{aq})$. After the reaction ran to completion, there were 1.50 mol Cl^- ions present in solution. How many moles of MgCl_2 were reacted?
- 0.75 moles MgCl_2
 - 1.50 moles MgCl_2
 - 2.25 moles MgCl_2
 - 3.00 moles MgCl_2
8. Which of the following contains the most moles of I: a 10.00 g sample of KI, a 10.00 g sample of CaI_2 , or a 10.00 g sample of HIO_3 ? (M = 166 g/mol KI, 294 g/mol CaI_2 , 176 g/mol HIO_3)
- a 10.00 g sample of KI
 - a 10.00 g sample of CaI_2
 - a 10.00 g sample of HIO_3
 - They are all the same

For Questions 9–10, please consider the following balanced reaction:



9. How many total species are present in the net ionic equation for this reaction?
- 3
 - 5
 - 6
 - 8
10. If the reaction starts with 5.00 grams of K_2CO_3 and 5.00 grams of CaCl_2 , how many grams of CaCO_3 are yielded at completion? (M = 138.21 g/mol K_2CO_3 , 110.98 g/mol CaCl_2 , 100.09 g/mol CaCO_3)
- 1.81 g CaCO_3
 - 3.62 g CaCO_3
 - 6.94 g CaCO_3
 - 7.24 g CaCO_3

2. [24] CHEMICAL REACTIONS.

- a. Consider the following unbalanced reaction:



- i. [2] What are the oxidation and reduction half-reactions?
- ii. [4] Write the balanced redox reaction equation.
- iii. [4] If 23.00 grams of KMnO_4 are reacted with 17.00 grams of Na_2SO_3 in an excess of H_2O , how many grams of KOH are yielded?
- iv. In a few words, explain **how** and **why** the reaction above would shift if...
 - a) [2] Na_2SO_3 was removed?
 - b) [2] H_2O was added?
 - c) [2] MgSO_4 was added?

- b. For each of the following prompts:
- 1) Write the unbalanced chemical equation for the reaction, including state symbols.
 - 2) Balance the chemical equation.
 - 3) Identify the reaction type (synthesis, decomposition, single replacement, double replacement, combustion).
- i. [2] A strip of magnesium is burned in air.
- ii. [2] Solutions of silver nitrate and aluminum chloride are mixed.
- iii. [2] Powdered potassium bicarbonate is added to aqueous acetic acid.
- iv. [2] Hydrogen chloride gas is bubbled through water.